🤖 Social Media Sentiment Analysis using Machine Learning

Analyze and classify social media text into **Positive**, **Negative**, or **Neutral** sentiments using Machine Learning and Natural Language Processing (NLP) techniques.

This project demonstrates an end-to-end NLP pipeline including:

- Data preprocessing
- Text vectorization
- Model training
- Sentiment prediction
- Performance evaluation
- Data visualization

📌 Project Overview

Social media platforms generate massive amounts of textual data every day. Understanding user sentiment helps businesses and organizations:

- monitor public opinion
- analyze customer feedback
- track brand reputation
- improve products and services

This project applies Machine Learning algorithms to perform sentiment analysis on social media text data and classify emotions expressed in posts/comments/tweets.

🚀 Features

✅ Text preprocessing and cleaning
✅ Stopword removal and tokenization
✅ TF-IDF / Count Vectorization
✅ Multiple ML models for comparison
✅ Sentiment prediction system
✅ Accuracy and evaluation metrics
✅ Data visualization and insights
✅ Easy-to-use modular structure

🧠 Machine Learning Models Used

- Logistic Regression
- Naive Bayes
- Support Vector Machine (SVM)
- Random Forest Classifier

🛠️ Technologies Used

Technology	Purpose
Python	Core programming
Pandas	Data handling
NumPy	Numerical operations
Scikit-learn	Machine learning
NLTK	NLP preprocessing
Matplotlib	Data visualization
Seaborn	Statistical plots
WordCloud	Word cloud generation
Jupyter Notebook	Development environment

📂 Project Structure

```
Sentiment-Analysis/
│
├── data/
│   └── dataset.csv
│
├── notebooks/
│   └── sentiment_analysis.ipynb
│
├── src/
│   ├── preprocessing.py
│   ├── train_model.py
│   ├── predict.py
│   └── visualization.py
```

```
│
├── outputs/
│   ├── confusion_matrix.png
│   ├── sentiment_distribution.png
│   └── wordcloud.png
│
├── models/
│   └── sentiment_model.pkl
│
├── requirements.txt
├── README.md
└── app.py
```

📊 Dataset

The dataset contains social media posts/comments labeled with sentiment classes such as:

- Positive 😊
- Negative 😠
- Neutral 😐

Example:

Text	Sentiment
"This product is amazing!"	Positive
"Worst experience ever."	Negative
"The service was okay."	Neutral

⚙️ Data Preprocessing

The following preprocessing steps were applied:

- Lowercasing text
- Removing punctuation
- Removing stopwords
- Tokenization
- Lemmatization/Stemming
- Removing URLs and special characters

📈 Model Performance

Model	Accuracy
Logistic Regression	XX%
Naive Bayes	XX%
SVM	XX%
Random Forest	XX%

> Replace the above values with your actual results.

📉 Visualizations

The project includes:

- Sentiment distribution graphs
- Confusion matrix
- Word clouds
- Accuracy comparison charts

▶️ How to Run the Project

1️⃣ Clone the Repository

```
git clone https://github.com/your-username/social-media-sentiment-analysis.git
cd social-media-sentiment-analysis
```

2️⃣ Install Dependencies

```
pip install -r requirements.txt
```

3️⃣ Run the Notebook

```
jupyter notebook
```

OR run the application:

```
python app.py
```

💡 Example Prediction

```
text = "I really love this product!"
prediction = model.predict([text])

print(prediction)
```

Output:

```
Positive
```

🔍 Future Improvements

- Deploy using Streamlit or Flask
- Add Deep Learning models (LSTM/BERT)
- Real-time Twitter sentiment analysis
- Multilingual sentiment analysis
- Live dashboard and analytics
- Emotion detection beyond polarity

🌐 Possible Applications

- Brand monitoring
- Customer feedback analysis
- Product review analysis
- Political sentiment tracking
- Social media analytics

- Market research

🤝 Contributing

Contributions are welcome!

If you'd like to improve this project: 1. Fork the repository 2. Create a new branch 3. Commit your changes 4. Submit a pull request

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📜 License

This project is licensed under the MIT License.

👨‍💻 Author

Developed by Sai Teja

If you found this project useful, feel free to ⭐ the repository.